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Dialysis vintage and outcomes in renal transplantation

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REVIEW
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ABSTRACT

Pre-emptive kidney transplantation is the recommended strategy for patients with end-stage renal failure in all guidelines [Kidney Disease: Improving Global Outcomes (KDIGO), The Australian and New Zealand Dialysis and Transplantation Registry (ANZDATA), European Renal Best Practice Guideline (ERBP), British Transplant Society (BTS)]. This recommendation is intuitive and based on few older studies with considerable limitations. In addition, there is conflicting evidence as to whether the duration of dialysis vintage impacts on graft and patient survival after transplantation. The objective of this structured review was to critically review the published evidence on dialysis vintage and outcomes by including the most recent papers on that topic. We searched Medline using keywords for kidney transplantation, pre-emptive, dialysis vintage and relevant outcomes, and found 14 eligible cohort studies. The best evidence was found for pre-emptive transplantation, which was found to be associated with a lower risk of actual graft loss (including death as event) compared with non-pre-emptive transplantation. When only patients were considered that have been registered pre-emptively but then received or did not receive a pre-emptive transplant, the association with functional graft survival (excluding death as event) was only marginal. Dialysis vintage had a graded association with patient survival in most of the studies, but an unclear estimate with functional graft survival. Older studies also found an association of dialysis vintage with death-censored graft survival, but this association is likely confounded by selection and the competing risk of death and was no longer observed in recent eras, i.e. in transplants performed in the last decade. In summary, the recommendation for pre-emptive kidney transplantation for optimal patient and graft survival remains valid even in recent periods but the association of dialysis vintage after dialysis initiation with death-censored graft survival is less clear. The association of dialysis vintage with mortality after transplantation depends on the median duration of dialysis of the wait-listed population as well as acceptance rates for transplantation, and may thus be country specific. Nevertheless, it is reasonable to advocate pre-emptive kidney transplantation in all age groups. What remains unsolved is the selection issues since the reasons for longer waiting time on dialysis are difficult to capture in retrospective observational studies, and lead time as well as immortal time bias may have confounded the mortality data.

Keywords: dialysis, end-stage renal disease, epidemiology, survival analysis, transplantation

INTRODUCTION

Almost every article on renal transplantation starts with the statement that renal transplantation is the treatment of choice for end-stage renal disease (ESRD). Transplantation provides a higher quality of life than dialysis and is cost-effective in the long run [1]. Although this is likely correct for the majority of patients with ESRD, there is little evidence supporting this claim for special categories of patients such as the elderly, because most of the studies are limited by selection bias as well as immortal time bias [2]. Additionally, there is not a single study in even the most extremes of advanced age patients that shows the opposite, i.e. a detrimental effect of transplantation, simply because unfit patients on the waiting list are delisted permanently or temporarily and thus will never or only after prolonged waiting time be transplanted. However, the patients remaining on dialysis are the majority, as only roughly 15–20% of all prevalent dialysis patients are considered fit for wait listing. The time to equal survival after transplantation compared with wait-listed patients on dialysis is >1 year in patients above 60 years of age [3] However, if selected super survivors with dialysis vintage of >10 years receive a renal allograft, it is still better compared with remaining wait listed on dialysis [4]. On the other hand, in the USA at least half of the patients aged >60 years placed on the waiting list will die before a transplant is available [5]. So, the intuitive consequence of these reports is optimal individual patient selection for better balancing the thin edge between futility and efficacy in the elderly transplant candidate.

This brief introduction of articles with limitations and caveats makes it clear that one needs to be careful not to confuse statistical associations observed in cohort studies with causal inference which, however, would be needed to support

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Table 1. Evaluation of study characteristics and outcomes

Study number	LD	DD	Pre-emptive versus HD vintage	HD vintage graded	Functional graft survival	Actual graft survival	Patient survival	Limitations (era effect, selection bias, lead time bias, immortal time bias)	Contributing factors (national regulations, SES)
Wolfe *et al.* [3]		X		↑			↑	EE, SB, LTB, IB	USA, SES
Meier-Kriesche *et al.* [11]	X	X	X	↑	↑	↑	↑	EE, SB, LTB, IB	USA, SES
Mange *et al.* [9]	X	X	X		↑	↑	↑	EE, SB, LTB, IB	USA, SES
Kasiske *et al.* [10]	X	X	X		↑	↑	↑	EE, SB, LTB, IB	USA, SES
Meier-Kriesche and Kaplan [17]	X	X	X	↑	↑	↑	↑		USA
Goldfarb-Rumyantzev *et al.* [19]	X	X	X	<6 months versus pre-emptive—no effects	↑	↑	↑	EE, SB, LTB, IB	USA
Becker *et al.* [12]	X	X	X		↑↔			EE (less effect in recent eras)	USA
Augustine *et al.* [20]	X	X	X ($n = 8$)	↑ acute rejection					USA
Johnston *et al.* [14]	X	X	X	↑	↑	↑	↑	SB, LTB, IB	USA
Schold [15]	X	X	X				↑	SB, LTB, IB	USA
Abramowicz *et al.* [8]	X		X	↑ but <12 months versus pre-emptive—no effects	↑	↑	↑		Meta-analysis
Haller *et al.* [13]	X	X	X	↑ but <18 months versus pre-emptive—no effects	↑ ↔ (no graded effect > year 2000)	↑	↑	SB, LTB, IB	EU (AUT)
Rose *et al.* [4]		X		>10-year dialysis versus HD			↑ (after 675 days)	SB, LTB, IB	USA (SRTR)
Prezelin-Reydit *et al.* [16]	X	X	X	X	↑	↑	↑	SB, LTB, IB	French DB

AUT, Austria; DD, deceased donor; EE, era effect; EU, European Union; DB, database; HD vintage, haemodialysis vintage; IB, immortal time bias; LD, living donor; LTB, lead time bias; SB, selection bias; SES, socioeconomic status; X, refers to whether live/deceased donors were included

recommendations with confidence. However, it is simply impossible to randomize patients to kidney transplantation or remaining on dialysis as organ availability determines waiting time. In our opinion, the equipoise principle would not be violated, especially in the very elderly, simply because of profound limitations and uncertainty in the currently existing non-randomized evidence base. It has happened in other clinical areas that a clinical routine suggested a best care procedure without evidence and which was swept away by randomized controlled trials that were initially deemed to be unethical [6]. Even well-known ethicists have published influential editorials advocating informativeness and ethical correctness of sham operation studies to elicit unclear practice patterns and their advantage over continuing with unproven procedures [7].

Therefore, the clinically important topic of dialysis vintage impact on kidney transplant outcome requires an in-depth compound summary and interpretation of the available observational data. We therefore searched for all relevant cohort studies investigating the impact of dialysis vintage on transplant outcomes in Medline from inception until 8 February 2018 using a Boolean combination of the following keywords: 'kidney transplantation', 'dialysis vintage', 'pre-transplant dialysis', 'dialysis duration', 'preemptive', 'graft' or 'transplant survival' or 'loss', 'mortality', 'death' and 'patient survival'. The search yielded 414 citations, and 14 were eligible for inclusion in our structured literature review. Table 1 lists all relevant articles on the topic of dialysis vintage and post-transplant outcomes in chronological order. The related topic of pre-emptive live donor transplant has been addressed recently in this journal in a position statement and systematic review from the DESCARTES working group and European Renal Best Practice Guideline (ERBP) of the ERA-EDTA [8].

PRE-EMPTIVE KIDNEY TRANSPLANTATION VERSUS HAEMODIALYSIS VINTAGE

The largest evidence base exists for comparing pre-emptive kidney transplantation with wait-listed patients on dialysis. Irrespective of whether deceased or live donor transplants were analysed, pre-emptive transplantation was always associated with a lower risk of graft failure and mortality.

Pre-emptive living-donor kidney transplantation

Abramowicz *et al.* recently conducted a systematic review on pre-emptive live donor kidney transplantation [8]. The authors retrieved 29 retrospective observational cohort studies (including three abstracts) that have been published after 1990. Most studies were at high risk of bias and adequate controlling of potential confounders was rare. Nevertheless, the uniform finding in all studies was that pre-emptive transplantation from a live donor was associated with a lower risk of functional graft failure and mortality. In detail, the risk of graft loss was reported to be lower in 13 of 23 studies and the risk of mortality was lower in 9 of 19 studies reporting these endpoints.

One of the first to study the association of pre-emptive live donor transplantation on outcomes was Mange *et al.* [9]. The investigators made use of 8481 live donor recipients in the USRDS in the years 1994–97 who either accumulated some dialysis vintage or were transplanted pre-emptively. They found

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that the risk of graft loss within the first few years was lower in the pre-emptive group, which was explained by the lower incidence of biopsy-confirmed rejection in this group because longer dialysis was associated with a higher degree of sensitization. Mortality data were not provided in detail but seemed to be at least numerically lower in the pre-emptive group.

Given the fact that the availability of a live donor represents *per se* a selection, it is of interest to decipher whether the observed association depends primarily on the donor organ source or other factors. The logical approach is to compare outcomes between pre-emptive live and deceased donor transplantation.

Pre-emptive deceased and live donor kidney transplantation

In the early 2000s it was still unclear whether pre-emptive transplantation should be recommended as quantitative data were still scarce. Kasiske *et al.* were among the first to analyse the USRDS registry in order to address this lack of information [10]. The uniform finding was that both live donor as well as deceased donor pre-emptive kidney transplantation were associated with a lower risk of graft loss and mortality compared with non-pre-emptively transplanted patients. However, these findings must be interpreted cautiously as almost 25% of live donor transplants were performed pre-emptively whereas this applied to only 7% of the deceased donor grafts. Clearly, this fact uncovers the heterogeneity of the two allograft recipient populations, although the reported relative effect sizes within each group are remarkably similar. The hazard ratio (HR) of death comparing pre-emptive with non-pre-emptive transplantation for live donor graft recipients was 0.69 [95% confidence interval (CI) 0.56–0.85] and 0.84 (95% CI 0.71–0.99) for deceased donor transplants, respectively. The authors summarized dialysis vintage as a dichotomous variable and thus the effect of dialysis vintage in the USRDS population can only be estimated from the Meier-Kriesche article [11].

A few years later, pre-emptive transplantation was evaluated in the subgroup of patients with diabetes Types 1 and 2 as well as simultaneous pancreas and kidney transplants [12]. The investigators made use of the Scientific Registry of Transplant Recipients database (SRTR), and in line with previous studies they found a reduced risk for graft failure and mortality in pre-emptively transplanted patients. Interestingly the association with graft failure, however, lost statistical significance in pre-emptive deceased donor transplants in recent eras. This observation has also been reported later by Haller *et al.* in an Austrian population that included only roughly 20% of patients with diabetes [13]. The era effect seems to play a key role in patient selection and outcomes interpretation. For this review we reanalysed the data to evaluate whether older age, i.e. >60 years, is an effect modifier for the association of pre-emptive transplantation benefits. We did not observe an interaction of age and pre-emptive transplantation, suggesting older age should not be restricted from pre-emptive transplantation.

A very interesting study was conducted by a group from Vancouver using USRDS data. Johnston *et al.* [14] addressed the question of whether a pre-emptive second kidney transplantation provides any benefit compared with some time on dialysis between the failed first graft and the second transplant. Roughly 60% of the pre-emptive grafts came from live donors, whereas only 30% had a live donor in the post-dialysis vintage group. No formal analyses of effect modification of donor source in the re-transplant patients have been conducted. It may be assumed, however, that no significant interaction occurred since early mortality after engraftment should not be different between deceased and live donor organs. The main message of this article was that findings from the pre-emptive first transplant literature are not directly generalizable to the second graft. In fact, the benefit of a pre-emptive second transplant seems primarily explained by a lower risk of mortality rather than a difference in functional graft survival. Clearly, the risk of graft failure for the second graft depends on the success of the first transplant. If the first transplant failed within the first year after engraftment, then the risk for premature graft failure for the second graft was likewise high.

Besides, dialysis vintage seems to be a key predictor of outcomes in transplant centre performance. Schold *et al.* studied the centre performance effect on outcomes [15]. The reassuring message is that even in the lowest performing centres, the benefit of a kidney transplant from a deceased donor outweighs the mortality risk on dialysis. Nevertheless, the magnitude of the survival benefit depended on centre performance.

Recently, a group from Bordeaux analysed the French national transplant registry to estimate the association of pre-emptive deceased and live donor transplantation on graft and patient survival as well as to compare mortality with wait-listed patients still on dialysis and non-pre-emptively transplanted patients [16]. The authors found a more pronounced difference between live and deceased donor pre-emptive transplantation, i.e. the HR of actual graft loss comparing pre-emptive and non-pre-emptive transplantation was 0.32 (95% CI 0.19–0.55) and 0.59 (95% CI 0.53–0.64), respectively. This association remained significant even for short duration of dialysis before transplantation. In a subgroup analysis of patients who had been wait listed before initiation of renal replacement therapy, the association was still present but, importantly, the effect size was much lower, i.e. the HR of actual graft loss was 0.70 (95% CI 0.57–0.86). This is a good indicator that the benefits of pre-emptive transplantation are overestimated in comparisons with all wait-listed patients. The risk of mortality was also found to be lower in pre-emptively transplanted patients compared with transplants following waiting time on dialysis (HR 0.60, 95% CI 0.50–0.71). In accordance to the aforementioned findings on graft survival, the association was weaker in patients who have been wait listed pre-emptively. In fact, this group no longer had a reduced risk of mortality by pre-emptive transplantation although the parameter estimate pointed towards a lower risk (HR 0.87, 95% CI 0.62–1.23).

Haller *et al.* investigated the impact of dialysis vintage using the Austrian Dialysis and Transplant Registry [13]. This study precisely distinguished the three main outcomes in kidney transplantation, i.e. functional and actual graft survival as well as mortality with a functioning graft. All together 461 pre-emptive deceased (204) and live donor (257) transplants were

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considered and compared with 6979 first kidney transplant recipients with varying durations of dialysis vintage. In order to account for the previously observed era effect, the analysis was stratified for transplants performed before and after the year 2000. Dialysis vintage was categorized in tertiles, i.e. <1.5 and >3.1 years, the median follow-up after transplantation was 8.2 years. The key findings were that pre-emptive transplantation was associated with a lower risk of functional graft loss, but this effect was driven mainly by living donors and the era before the year 2000. Additionally, in patients without pre-emptive transplantation, the investigators observed no graded effect of dialysis vintage on functional graft survival but with mortality and, thus, also actual graft survival. This observation is in line with the French study, which also found a higher mortality rate with longer dialysis vintage before transplantation [16]. The authors also showed an increase risk of actual graft loss with increasing dialysis vintage but not for functional graft loss, and thus one may speculate based on their competing risk analysis that at least half of the events may be attributed to death.

DIALYSIS VINTAGE

In a very elegant approach, Meier-Kriesche and Kaplan investigated the contribution of dialysis vintage to graft and patient survival after transplantation by making use of paired deceased donor organs that were transplanted into recipients with different dialysis vintage duration [17]. One group of recipients with a dialysis vintage <6 months was compared to a group of recipients of one kidney from the same donor with a dialysis vintage of >2 years. The main finding was that short dialysis led to a better graft and patient survival. It was observed that 10 years after transplantation, 63% of patients with a short dialysis time had a functioning graft, but only 29% of patients with >2 years of dialysis before transplant. In a comparison of these data with outcomes following live donor transplantation in the USA, the authors found that graft survival of a deceased donor transplant in a patient with short dialysis vintage is equivalent to a live donor transplant of a recipient with >2 years of dialysis vintage. These findings are impressive but were unfortunately analysed only by univariable methods and, thus, neither confounding nor selection bias were addressed.

The same group used the USRDS registry shortly before to decipher the effect of graded duration of dialysis vintage on functional graft and patient survival [11]. Both studies used data from as early as 1988 when the median graft survival was significantly lower compared with nowadays or even after the year 2000 [18]. The uniform finding in line with the era analysis by Haller *et al.* was that dialysis vintage impacted on patient survival [13]. The discrepancy in functional graft survival may be partly explained by population admix and difference in socio-economic status as well as health insurance coverage beyond the third year. The era effect was quantified by Meier-Kriesche *et al.* [11] as an annual reduction of the hazard rate of functional graft loss by 4% with each increase of a calendar year. This is a very impressive improvement and may reflect the steep learning curve in the last decades of the last century.

Some years later Goldfarb-Rumyantzev *et al.* used the same database and almost same era of transplantation to elucidate whether a safe short dialysis vintage period exists that could facilitate planning of the transplant [19]. Dialysis vintage was very narrowly categorized into <2 weeks, <2 months, <6 months and then annually. Other than subsequent studies the authors found that a shorter dialysis period of up to 6 months had no impact on graft survival and a dialysis period of <1 year did not affect patient survival.

A group from Cleveland investigated the contribution of prolonged dialysis vintage to acute rejection and graft loss in a population with a high proportion of black patients [20]. The main finding was that dialysis time leads to an increase in T-cell alloimmunity, which may contribute to the inferior outcomes in this population with a longer dialysis vintage.

Some unfortunate patients with ESRD remain on the transplant waiting list for a long time, most commonly due to a higher rate of sensitization against Human Leukocte Antigen (HLA) antibodies, blood group 0 and comorbidities leading to temporary delisting. Until recently, it was unclear whether such patients with a dialysis vintage of >10 years would derive any benefit from a deceased donor kidney transplant. Rose *et al.* addressed this issue by analysing the USRDS database [4]. The authors were able to identify 2320 patients who received a transplant after 10 years of dialysis and compared these with wait-listed patients for >10 years but still on dialysis. Their main finding was that transplantation was associated with longer overall survival but it took on average almost 2 years until this survival benefit was statistically evident despite receiving good-quality kidneys.

LIMITATIONS AND CONTRIBUTING FACTORS

As outlined above, the evidence base on dialysis vintage at large arises exclusively from observational impeding causal inference for advocating treatment recommendations. Nevertheless, there is a great need to guide treatment decisions even in areas with no evidence from randomized clinical trials. In such situations, it is of utmost importance to not confuse causation with association and to take the higher risk of bias, and therefore higher uncertainty, into account.

Dialysis vintage, i.e. the time spent on dialysis prior to transplantation, is modified by a complex variety of factors introducing a broad spectrum of limitations for generalizability of the study findings reviewed above. First and foremost, the probably strongest limitation in the area of dialysis vintage and outcomes after transplantation is the selection of patients from wait lists. There is usually an underlying clinical reason to either decline or defer wait listing, and even the information about the possibility of a pre-emptive transplantation from either an available live donor or potentially a deceased donor in case of timely wait listing is heterogeneous provided throughout the world. Another important limitation is lead time bias of chronic kidney disease and comorbidities that certainly contribute to the post-transplant outcomes but are quantitatively difficult to capture.

Furthermore, the comparison of transplanted versus wait-listed patients introduce immortal time bias, suggesting that transplant patients must have lived long enough to receive the transplant but wait-listed patients may die on the waiting list

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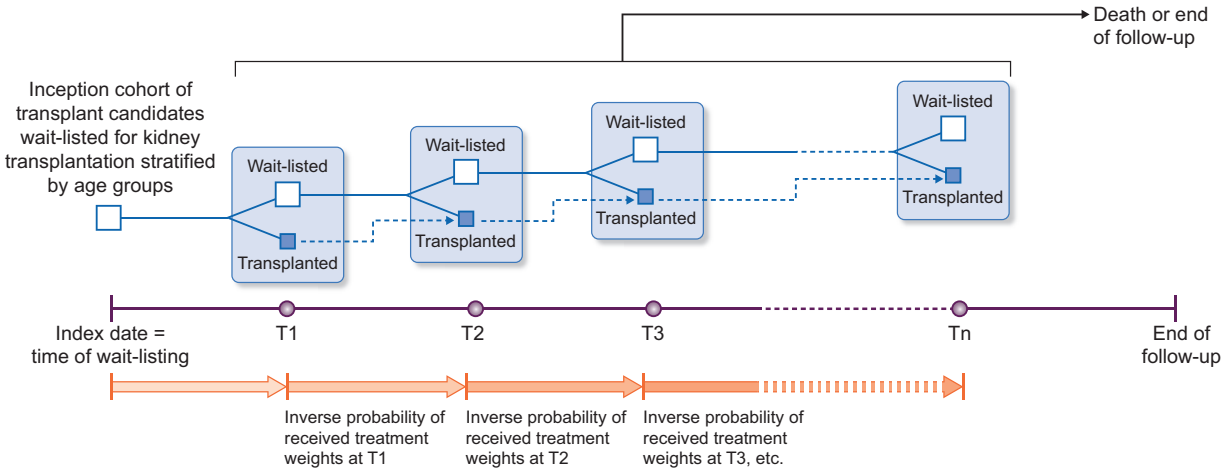

FIGURE 1: Illustration of the immortal time bias. Patients with a transplant always exhibit a survival benefit because the wait-listed patients had to live at least until a transplant became available. To address this issue a Cox model with pre-specified, fixed landmarks over time would be required as used by Haller *et al.* recently [22]. The idea is that at the end of one observation period, i.e. the landmark T1 or T1 to T2, etc., a pseudo-randomization by marginal structural models, for example, is applied to balance 'group assignment', i.e. the probability of receiving a graft.

and thus never have the chance to switch to the transplant group. The contemporary primary statistical solution to this intrinsic weakness is landmark modelling, which is very infrequently used in the transplant literature [21, 22] (Figure 1).

Another well-known phenomenon that requires attention is the competing risk situation of death and graft loss. An extreme hypothetical example would be that very elderly recipients received a pre-emptive transplantation with good quality organs and die shortly after transplantation with a functioning graft. This certainly cannot count as success but if only graft survival was considered it would be close to 100%. On top, the competing risk of death does not necessarily have to be procedurally related but might confound the overall interpretation. For example, many countries banned smoking in public places and offered smoking cessation programmes, which led to a rather swift reduction in cardiovascular deaths and a reduction in lung cancer on a population level. Thus, these national and temporal factors need to be taken into account when comparing country or even transcontinental data.

An important further specialty of transplantation is the non-uniformity of algorithms and eligibility criteria for transplant wait listing. Accordingly, the rate of live donor transplants around the world ranges from almost 0% to 100%. Intuitively pre-emptive kidney transplantation can be advocated in some regions but not in others.

It is even more complex due to different national regulations and legislation. For example, the waiting time for a deceased donor kidney in Germany is longer than 5 years and among the highest in Europe (Deutsche Stiftung Organtransplantation). At the same time, while some countries such as UK and the Netherlands derive almost half of the deceased donor organs from donors with cardiocirculatory death, Germany forbids the use of such organs by law. One can intuitively conclude from these different circumstances that the acceptance rate for wait listing must be different among countries, and it is not only driven by legislation but also population demographics. The median life expectancy among European countries differs by >10 years (http://ec.europa.eu/eurostat/statistics-explained/index.php/File:Life_expectancy_at_birth,_1980-2015_(years).png). Another important point to consider is the national and even state policies of dialysis provision. There are data showing that patients from for-profit dialysis providers are less likely to be wait listed in time compared with patients in non-for-profit units [23]. If waiting time on dialysis is longer and thus the risk of sensitization higher, a special programme for those patients needs to be available. While larger transplant consortia have programmes such as the Acceptable mismatch program (AMM), Eurotransplant Senior Program (ESP) and Kidney Paired Donation (KPD) in place, others do not. Such programmes allow to wait list and transplant also disadvantaged transplant candidates who otherwise will never have the chance for a functioning graft.

CONCLUSIONS

Although there are no randomized controlled trials proving that pre-emptive live and deceased donor kidney transplantation is the preferred treatment for eligible patients with ESRD it is reasonable to justify this statement with the currently available observational studies taking their limitations into account.

Recently, there is increasing uncertainty as to whether very elderly patients or patients with very long dialysis lead time actually benefit after transplantation. In clinical decision-making, physicians take many more parameters into account, which are unfortunately rarely reflected in large database studies. These parameters range from the socioeconomic status of patients to overall comorbidities, quality of life on dialysis and last, but not least, the median waiting time in the transplant centre. Ultimately, pre-emptive transplantation and wait listing for a transplant needs to be discussed with the patient considering all individual aspects. The soon to be published Kidney Disease:

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Improving Global Outcomes (KDIGO) guidelines on wait listing will provide further detailed information on that topic.

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CONFLICT OF INTEREST STATEMENT

None declared.

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